MATHS MANTHAN
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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 10** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 10: Circles** **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS (LA) questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal choices.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. If TP and TQ are two tangents drawn from T to a circle with centre O such that $\angle PTQ = 70°$, then $\angle OPQ$ is equal to: [1]

(A) $35°$ (B) $45°$ (C) $55°$ (D) $70°$

Q2. In a right triangle ABC with $\angle B = 90°$, $AB = 8$ cm and $BC = 6$ cm. The radius of the circle inscribed in the triangle is: [1]

(A) 1.5 cm (B) 2 cm (C) 2.5 cm (D) 3 cm

Q3. If a chord of the outer circle of two concentric circles has length 14 cm and is tangent to the inner circle of radius 24 cm, the outer radius is: [1]

(A) 25 cm (B) 26 cm (C) 28 cm (D) 31 cm

Q4. If PA and PB are tangents to a circle from P such that $PA = 9$ cm and $\angle APB = 60°$, then the length of chord AB is: [1]

(A) 4.5 cm (B) 9 cm (C) $9\sqrt{3}$ cm (D) 6 cm

Q5. The maximum number of parallel tangents that a single circle can have at most is: [1]

(A) 1 (B) 2 (C) 3 (D) Infinite

Q6. **Assertion (A):** If a right triangle ABC with sides 3 cm, 4 cm, 5 cm circumscribes a circle, then the radius of the incircle is 1 cm.
Reason (R): The inradius r of a right triangle with legs a, b and hypotenuse c is given by $r = \frac{a+b-c}{2}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. The tangent at a point C of a circle and a diameter AB when extended intersect at P. If $\angle PCA = 110°$, find $\angle CBA$. [2]

Q8. A circle is inscribed in $\triangle ABC$ touching AB, BC, CA at P, Q, R. If $AB = 12$ cm, $BC = 8$ cm, and $AC = 10$ cm, find the length of AD. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. From a point T outside a circle with centre O, tangents TP and TQ are drawn. If $OT = 13$ cm and radius is 5 cm, find the length of chord PQ. [3]

Q10. Prove that in two concentric circles, all chords of the larger circle which touch the smaller circle are of equal length. [3]

SECTION D: Long Answer / HOTS Type (2 × 5 = 10 Marks)

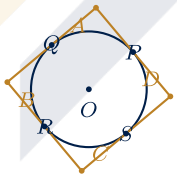
Q11. In the given figure, a circle touches all four sides of a quadrilateral $ABCD$ whose sides are AB, BC, CD, DA. (i) Prove that $AB + CD = AD + BC$. (ii) If $AB = 12$ cm, $BC = 15$ cm, $CD = 14$ cm, find DA. (iii) If $\angle B = 90°$ and $r = 4$ cm, find the lengths of the tangent segments on side AB. [5]

Q12. Two circles with centres O and O' of radii 5 cm and 3 cm touch each other internally. If the perpendicular bisector of the segment joining their centres meets the larger circle at points P and Q, find the length of chord PQ. [5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. Security Perimeter Wall Circumscribing a Vault

A security compound builds a four-sided boundary wall $ABCD$ circumscribing a circular bank vault. The wall lengths are $AB = 24$ m, $BC = 27$ m, $CD = 31$ m. Security sensors are positioned along the perimeter.



(i) State the theorem governing the side lengths of a circumscribed quadrilateral. [1]

(ii) Calculate the required length of the fourth boundary wall DA. [1]

(iii) Find the total perimeter of the security compound, OR if the vault has radius $r = 10$ m, find the total area of the four triangles formed by connecting the vault centre to the vertices. [2]

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